

C++2: Lesson 5 – Lab Project

1. Add **summary** at the top and **comments** where necessary.
- Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2 100 points Write a code that prompts the user to enter their name and age. 1. The name should contain only alphabetic characters (no numbers or special characters). 2. The age should be a positive integer. 3. If invalid input is detected: a. The respective function should throw an exception. b. The main() function should catch the exception, display an error message, and rethrow it for handling at a higher level. 4. After successful input validation, print the name and age. <table border="1"> <tr> <td data-bbox="159 793 293 867">Sample output</td> <td data-bbox="293 793 1549 989"> Name? Kim Jung Eun Age? 35 ----- Kim Jung Eun 35 </td> </tr> <tr> <td data-bbox="159 989 293 1066">File(s) to submit</td> <td data-bbox="293 989 1549 1066"> C205_2.cpp C205_2.png </td> </tr> </table>	Sample output	Name? Kim Jung Eun Age? 35 ----- Kim Jung Eun 35	File(s) to submit	C205_2.cpp C205_2.png	
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3 100 points Write a program that performs basic arithmetic operations using a class that can handle different types of data (e.g., integers, floating-point numbers). 1. Create a class that store two values (value1 and value2) of the same type. - It should include methods to perform addition and multiplication operations. - Use a constructor to initialize value1 and value2. 2. In the Test code: - Create two instances of the class: one for integers and another for floating-point numbers. - Perform addition and multiplication for both instances and display the results. 3. <table border="1"> <tr> <td data-bbox="159 1455 293 1528">Sample output</td> <td data-bbox="293 1455 1549 1766"> Enter two integer values: 10 5 Addition: 15 Multiplication: 50 Enter two double values: 12.3 8.2 Addition: 20.5 Multiplication: 98.1 </td> </tr> <tr> <td data-bbox="159 1766 293 1839">File(s) to submit</td> <td data-bbox="293 1766 1549 1839"> C205_3.cpp C205_3.png </td> </tr> </table>	Sample output	Enter two integer values: 10 5 Addition: 15 Multiplication: 50 Enter two double values: 12.3 8.2 Addition: 20.5 Multiplication: 98.1	File(s) to submit	C205_3.cpp C205_3.png	
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